Day 1-5 : Build and Configure a Small Office Network

Practical Assignment – Days 1 to 5

1. Objective

The objective of this practical was to build and configure a small office network using Cisco Packet Tracer. The practical involved static IP addressing, DHCP, DNS, HTTP, VLAN configuration, and Network Address Translation (NAT).

The main purpose was to understand how devices communicate within a network and how network services help manage connectivity.

2. Network Setup

The network was created using the following devices:

- 1 Router
- 1 Cisco 2960 Switch
- 5 PCs
- 1 Internal Server (DNS and HTTP)
- 1 External Server

Internal Network Details:

- Network Address: 192.168.10.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- DNS Server: 192.168.10.2

External Network Details:

- Network Address: 203.0.113.0/24
- Router External IP: 203.0.113.1
- External Server IP: 203.0.113.2

Task 1: Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.Objective

To assign static IP addresses to five devices in the same network and verify connectivity using the ping command.

Configuration

The following static IP addresses were assigned to the PCs:

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.10.12	255.255.255.0	192.168.10.1
PC3	192.168.10.13	255.255.255.0	192.168.10.1
PC4	192.168.10.14	255.255.255.0	192.168.10.1

Procedure

1. Opened each PC in Cisco Packet Tracer.
2. Navigated to Desktop → IP Configuration.
3. Selected Static configuration.
4. Entered the IP address, subnet mask, and default gateway.
5. Opened the Command Prompt on PC0.
6. Used the ping command to test connectivity with the other PCs.

Command Used:

ping 192.168.10.11

ping 192.168.10.12

ping 192.168.10.13

ping 192.168.10.14

Result

All five PCs were configured with unique static IP addresses in the same subnet. Ping tests from PC0 to the other PCs were successful.

Task 2: Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Objective

To configure DHCP on the router so that network devices can automatically receive IP addresses.

Configuration

DHCP was configured on the router using the following commands:

enable

configure terminal

ip dhcp excluded-address 192.168.10.1 192.168.10.99

ip dhcp pool OFFICE-LAN

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 8.8.8.8

exit

Explanation

- ip dhcp excluded-address reserves addresses from .1 to .99 so DHCP does not assign them.
- ip dhcp pool OFFICE-LAN creates a DHCP address pool.
- network defines the network from which addresses are assigned.
- default-router specifies the gateway.
- dns-server specifies the DNS server address.

PC0, PC1, and PC2 were changed to DHCP mode through Desktop → IP Configuration.

The router assigned unique IP addresses starting from 192.168.10.100.

Verification Command:

show ip dhcp binding

Result

DHCP successfully assigned IP addresses automatically to at least three PCs. The assigned addresses were unique, and the devices were able to communicate within the network.

Task 3: Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Objective

To configure a DNS server with a custom hostname and access a locally hosted webpage using that hostname.

Server Configuration

An additional Server-PT device was connected to the switch.

The following static IP configuration was applied:

Setting	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	192.168.10.2

DNS Configuration

1. Opened the server and navigated to Services → DNS.
2. Enabled the DNS service.
3. Added the following DNS record:

Field	Value
Name	musicplayer.local
Type	A Record
Address	192.168.10.2

The DHCP pool was also updated to provide 192.168.10.2 as the DNS server address.

PC0 was renewed using:

ipconfig /release

ipconfig /renew

HTTP Configuration

1. Opened Services → HTTP on the server.
2. Enabled the HTTP service.
3. Edited the index.html file.
4. Added a welcome message to the webpage.

Testing

Opened the web browser on a PC and entered:

http://musicplayer.local

Result

The custom hostname successfully resolved to the server's IP address, and the locally hosted welcome webpage opened in the browser.

This confirmed that DNS name resolution and HTTP services were working correctly.

Task 4: Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Objective

To create two separate VLANs and verify communication within the same VLAN while restricting communication between different VLANs.

VLAN Creation

The Cisco 2960 switch was configured with two VLANs:

enable

configure terminal

vlan 10

name Workstations

exit

vlan 20

name Smart-Devices

exit

Port Assignment

The following switch ports were assigned to the VLANs:

Switch Port	Connected Device	VLAN
Fa0/1	Router	VLAN 10
Fa0/2	PC0	VLAN 10
Fa0/3	PC2	VLAN 20
Fa0/4	PC1	VLAN 10
Fa0/5	PC3	VLAN 20

Fa0/6	PC4	VLAN 10
Fa0/7	DNS/HTTP Server	VLAN 10

Port Configuration

VLAN 10:

interface range fa0/1-2

switchport mode access

switchport access vlan 10

exit

interface fa0/4

switchport mode access

switchport access vlan 10

exit

interface range fa0/6-7

switchport mode access

switchport access vlan 10

exit

VLAN 20:

interface fa0/3

switchport mode access

switchport access vlan 20

exit

interface fa0/5

switchport mode access

switchport access vlan 20

exit

IP Configuration for VLAN 20

The following static IP addresses were assigned:

Device	IP Address	Subnet Mask
PC2	192.168.20.12	255.255.255.0
PC3	192.168.20.13	255.255.255.0

Testing

- Ping from PC2 to PC3 was successful.
- Ping from PC2 to PC0 was unsuccessful.

Result

Both VLANs were created successfully. Devices within VLAN 20 could communicate with each other, while communication between VLAN 10 and VLAN 20 was restricted.

This demonstrated network segmentation using VLANs. Inter-VLAN routing was not configured in this practical.

Task 5: Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Objective

To configure Network Address Translation (NAT) on the router and simulate external network access using an external server.

External Server Configuration

An additional Server-PT device was connected to the router's external interface.

The following IP configuration was applied:

Setting	Value
IP Address	203.0.113.2
Subnet Mask	255.255.255.0
Default Gateway	203.0.113.1

Router Interface Configuration

The router used two FastEthernet interfaces:

- FastEthernet0/0: Internal LAN interface.
- FastEthernet0/1: External network interface.

The NAT inside and outside interfaces were configured:

```
enable
configure terminal

interface fastEthernet 0/0
ip nat inside
exit

interface fastEthernet 0/1
```

ip nat outside

exit

NAT Access List Configuration

An access list was created to allow the internal network to use NAT:

access-list 1 permit 192.168.10.0 0.0.0.255

NAT Overload Configuration

The following command enabled NAT Overload (PAT):

ip nat inside source list 1 interface fastEthernet 0/1 overload

This allowed multiple internal devices to share the router's external IP address using different port or identifier mappings.

Saving Configuration

end

write memory

Testing

From PC0, the following command was executed:

ping 203.0.113.2

The NAT translation table was verified using:

show ip nat translations

The output showed the following translation:

NAT Field	IP Address
Inside Local	192.168.10.12
Inside Global	203.0.113.1
Outside Local	203.0.113.2
Outside Global	203.0.113.2

Result

NAT Overload was successfully configured. The router translated the internal private IP address into its external IP address.

The NAT translation table confirmed that traffic from the internal network was being translated when communicating with the external server.

The external server simulated an external network destination; this was not a connection to the real public Internet.